Compound vs Isolated Cyclone Shocks

Do Repeated Shocks Compound the Damage?

1. Motivation and Data

Question

Do cyclone shocks that follow closely after a previous shock — **compound shocks** — cause more or less damage than **isolated shocks**?

Two competing hypotheses:

- **Compounding damage**: A second shock hits while the economy is still recovering, amplifying the GDP loss (weakened infrastructure, depleted reserves)
- **Diminishing marginal damage**: The first shock already destroyed vulnerable capital, so a second shock has less left to destroy; or governments pre-position response capacity after the first event

Definition: A shock in year t is *compound* if there was also a p95 cyclone shock in $t-1$. Otherwise it is *isolated*.

The Threshold Problem

The baseline p95 threshold is computed across *all* country-years (88% of which have zero wind speed), yielding a threshold of just **60 knots** — barely tropical storm strength.

This means large cyclone-prone countries are "treated" nearly every year:

Country	**p95 (60 kts)**	**Cat 1+ (64)**	**Cat 2+ (83)**	**Cat 3+ (96)**
Australia	41 / 46 yrs	40	25	14
China	44 / 46 yrs	41	13	7
Philippines	42 / 46 yrs	39	30	17
Japan	40 / 46 yrs	37	11	4

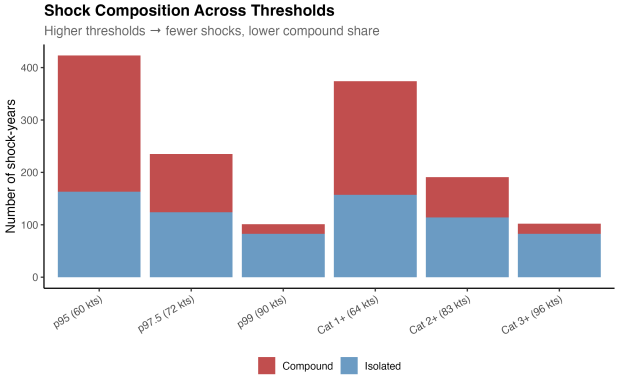
- At p95, these countries have $\geq 90\%$ compound shocks $\rightarrow$ no within-country variation
- Higher thresholds (Cat 2+, Cat 3+) capture only severe events and produce more isolated shocks

Shock Classification (Baseline p95)

	Count	**Share**
Total shock-years	423	100%
Compound (strict: shock in $t-1$)	260	61.5%
Isolated (no shock in $t-1$)	163	38.5%
Compound (broad: shock in $t-1$ or $t-2$)	311	73.5%
Isolated (broad: no shock in $t-1$ or $t-2$)	112	26.5%

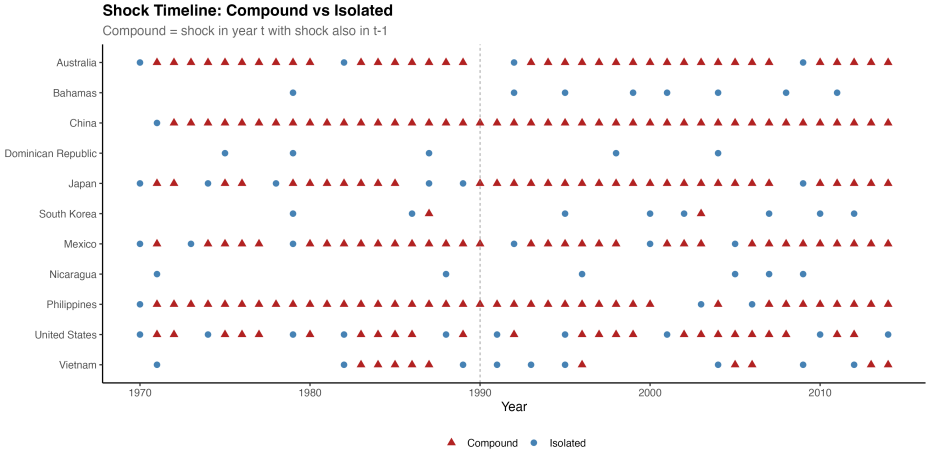
- Among the 40 ever-treated countries, nearly two-thirds of all shock-years are compound
- Compound share is stable across periods: 62.6% pre-1990, 60.6% post-1990
- 19 countries have *both* compound and isolated shocks $\rightarrow$ within-country variation exists

Shock Composition Across Thresholds



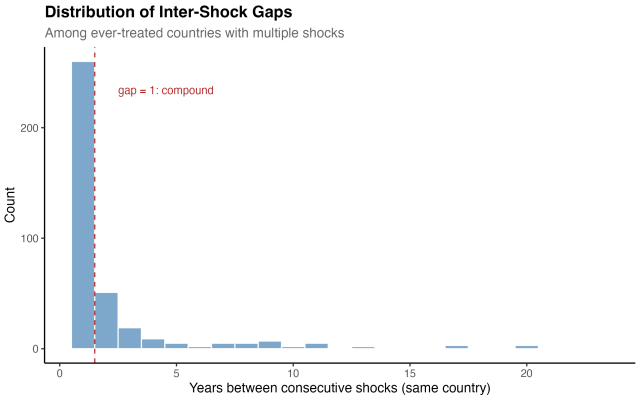
Higher thresholds sharply reduce both total shocks and the compound share: from 61.5% (p95) to 18% (Cat 3+). At Cat 2+ (83 knots), there are 191 shocks with a 40% compound share — a reasonable balance of power and variation.

Shock Timeline: Selected Countries



Large cyclone-prone countries (CHN, PHL, AUS, JPN, MEX) are hit nearly every year → almost all compound. Smaller Caribbean countries (BHS, NIC, DOM) have sporadic, isolated shocks.

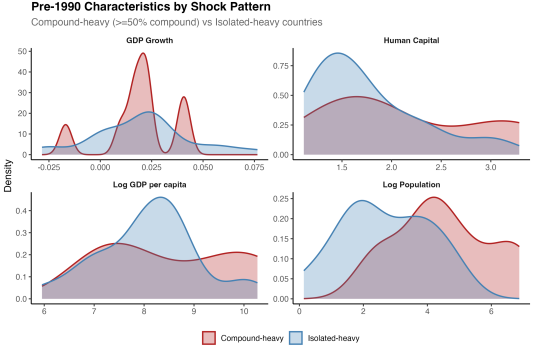
Inter-Shock Gap Distribution



The modal gap between consecutive shocks within a country is **1 year**. Most multi-shock countries experience near-continuous exposure, not discrete, well-separated events.

Identification Challenge

Compound and isolated shocks are **not randomly assigned**:



Compound-heavy countries are larger (log pop 4.5 vs 2.7), slightly richer, and have higher human capital. These are the large, frequently-hit economies (CHN, JPN, USA). Isolated-heavy countries are smaller Caribbean/Central American nations.

Identification Strategy

We address the selection problem through:

1. **Within-country variation**: Country FE + country-specific trends ensure we compare compound vs isolated shocks *within the same country's trajectory*
2. **Multiple FE specifications**: Year FE, region×year FE
3. **Controls**: Lagged GDP growth and lagged shocks absorb autoregressive dynamics
4. **Robustness**: Dropping always-compound countries, alternative definitions, continuous intensity

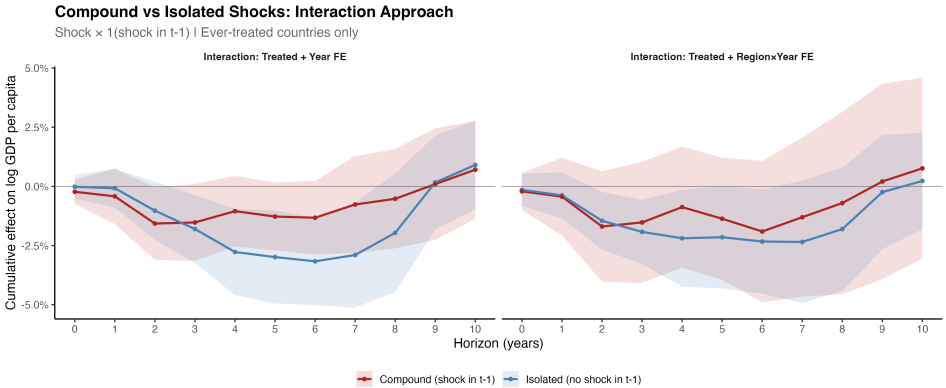
LP specification:

$$y_{i,t+h} - y_{i,t-1} = \beta_{\text{iso}} \cdot \mathit{Isolated}_{it} + \beta_{\text{comp}} \cdot \mathit{Compound}_{it} + \gamma' X_{it} + \alpha_i(t) + \delta_t + \varepsilon_{it}$$

where $\alpha_i(t)$ includes country-specific linear and quadratic trends.

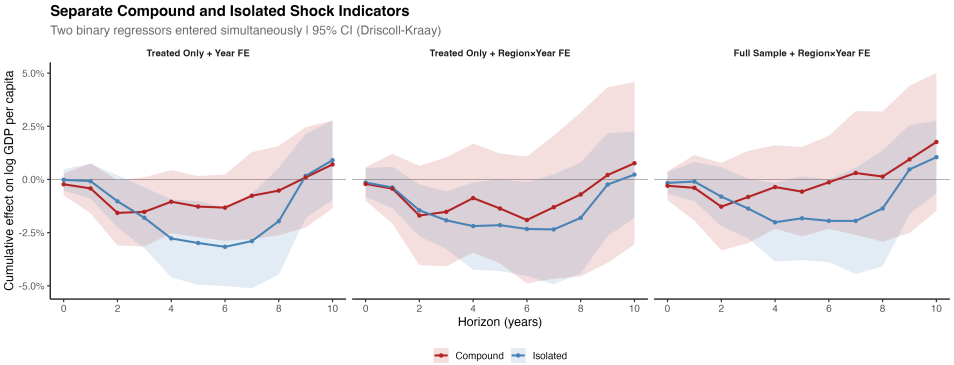
2. Main Results

Main Result: Interaction Approach



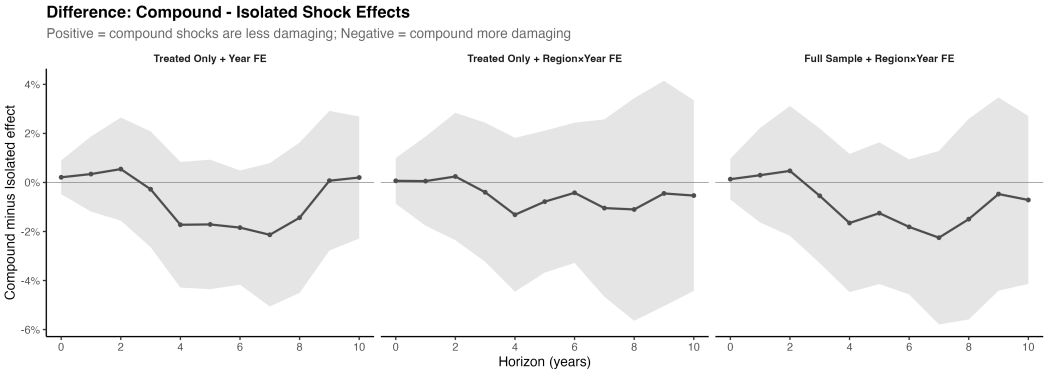
Isolated shocks cause a **persistent 3.0% GDP decline** at $h = 5$. Compound shocks cause only a **1.3% decline** (insignificant). The difference is not statistically significant ($p > 0.10$), but the point estimates diverge from $h = 2$ onward.

Separate Shock Indicators Across FE Specifications



The pattern is robust to FE specification: isolated shocks are more damaging across all three. Adding never-treated countries as controls (right) or switching to region×year FE (center) does not change the qualitative result.

Compound − Isolated Difference



The difference (compound − isolated) is consistently positive but **never statistically significant** at 95%. The wide CIs reflect limited identifying variation: only 19 countries have both types, and the country-specific trends absorb much of the signal.

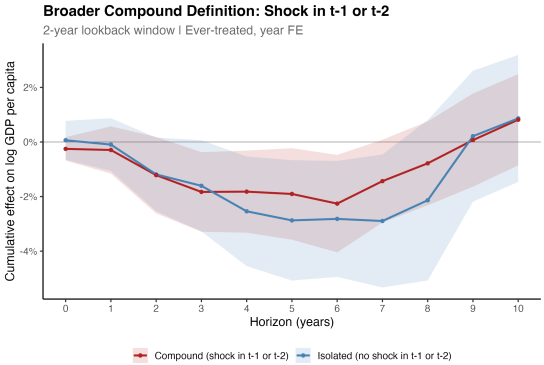
Results at $h = 5$

Specification	**Isolated**	**Compound**	**Diff**
Treated + Year FE	$-3.0\%^{**}$	-1.3%	$+1.7\%$
Treated + Region×Year FE	$-2.1\%^{*}$	-1.4%	$+0.8\%$
Full Sample + Region×Year FE	$-1.8\%^{*}$	-0.6%	$+1.3\%$

- Isolated shocks consistently larger in magnitude
- Compound shocks are smaller and often insignificant
- The difference is not statistically significant in any specification
- **Interpretation**: Compound shocks appear *less* damaging than isolated ones — consistent with diminishing marginal damage, not compounding

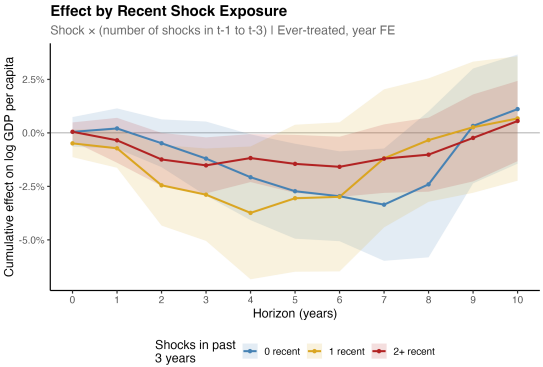
3. Robustness

Broader Compound Definition: Shock in $t-1$ or $t-2$



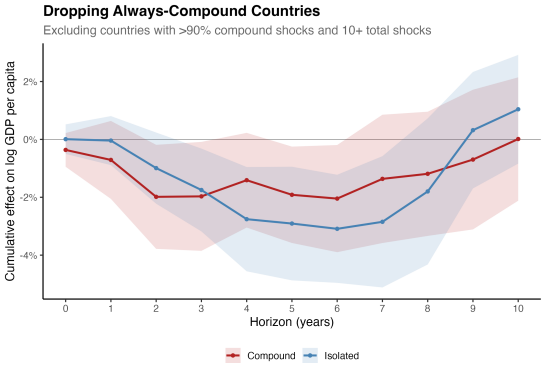
Extending the lookback window to 2 years: the pattern is similar. Isolated shocks (no shock in $t-1$ or $t-2$) are more damaging, but the gap narrows because some previously-isolated shocks are reclassified as compound.

Cumulative Exposure: 0, 1, or 2+ Shocks in Past 3 Years



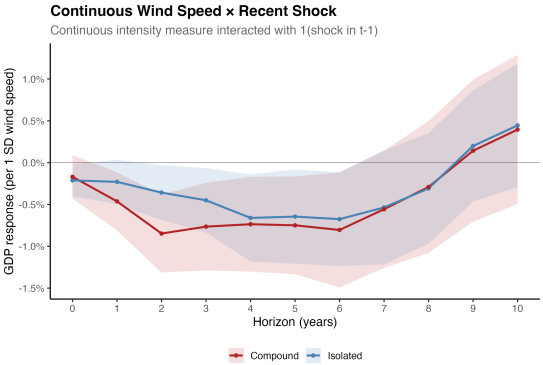
The dose-response is **non-monotonic**: 0 and 1 recent shocks produce similar damage (-2.7% and -3.1%), while 2+ recent shocks produce *less* damage (-1.4%). This is consistent with diminishing marginal effects at high exposure.

Dropping Always-Compound Countries



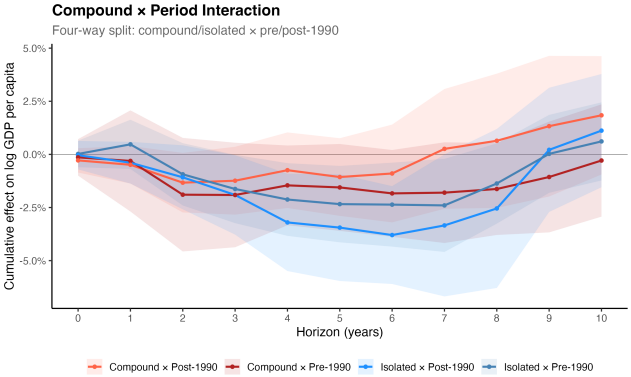
Dropping the 3 countries with $\geq 90\%$ compound shocks and ≥ 10 total shocks (CHN, PHL, AUS): the pattern persists. Isolated shocks: -2.9%; compound: -1.9%. This addresses the concern that the result is driven by these large, frequently-hit economies.

Continuous Wind Speed × Recent Shock



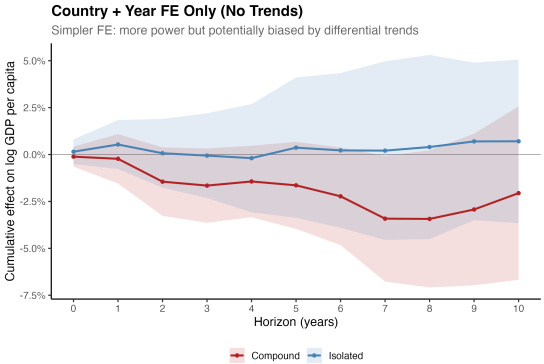
Using continuous wind speed (per 1 SD) instead of the binary indicator: the compound/isolated difference **disappears**. Both are $\approx -0.6\%$ to -0.7% per SD. The binary result may partly reflect intensity differences between compound and isolated shocks.

Compound × Period Interaction



Four-way split: Isolated post-1990 shocks are most damaging (-3.4%). Compound shocks are attenuated in both periods. The compound/isolated gap does not widen post-1990 — the period divergence in the main analysis operates through a different channel.

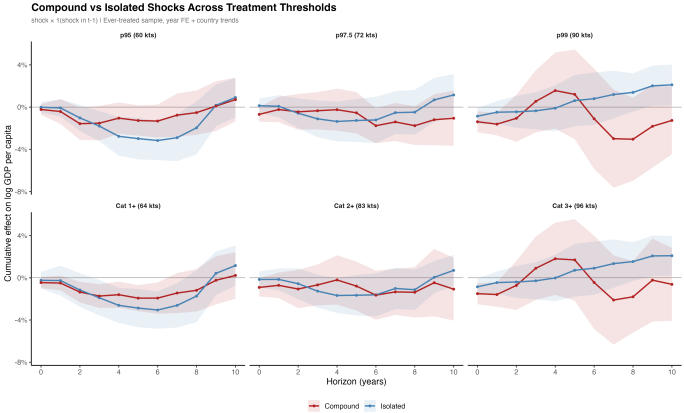
Country + Year FE Only (No Trends)



Without country-specific polynomial trends (more power, but potentially biased): the isolated effect vanishes and compound flips sign. Country trends are essential — without them, the LP conflates the shock effect with differential growth paths of frequently vs. infrequently hit countries.

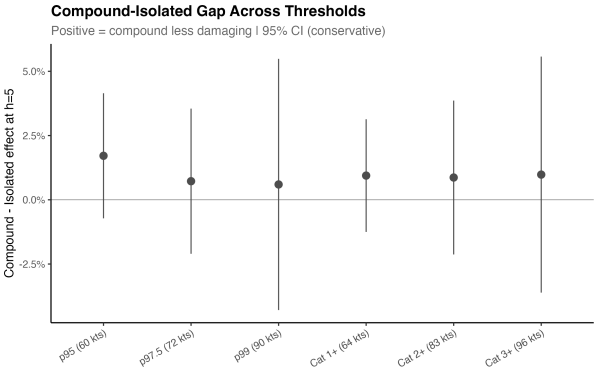
4. Varying the Treatment Threshold

IRFs Across Thresholds



As the threshold increases: (1) effects weaken and lose significance due to smaller samples; (2) at p99 and Cat 3+ ($\sim$100 shocks), both compound and isolated effects are near zero or positive. The compound/isolated gap is most visible at p95 and Cat 1+, where sample sizes are largest.

Compound−Isolated Gap Across Thresholds



The gap (compound − isolated) at $h = 5$ is **positive** at every threshold — compound shocks are consistently less damaging. But the gap is **never significant** and is largest at the lowest threshold where compound-heavy countries dominate.

Threshold Summary

Threshold	**Shocks**	**Treated**	**%Comp**	**Isol.** h=5	**Comp.** h=5
p95 (60 kts)	423	40	61.5%	−3.0%	−1.3%
p97.5 (72 kts)	235	29	47.2%	−1.3%	−0.5%
Cat 1+ (64 kts)	374	36	58.0%	−2.9%	−1.9%
Cat 2+ (83 kts)	191	28	40.3%	−1.7%	−0.8%
p99 (90 kts)	101	23	17.8%	+0.6%	+1.2%
Cat 3+ (96 kts)	102	23	18.6%	+0.7%	+1.7%

- At moderate thresholds (p95–Cat 2+): negative effects, compound less damaging
- At severe thresholds (p99, Cat 3+): effects flip positive — but with only ∼100 shocks, power is very limited and CIs are wide
- The compound/isolated pattern is qualitatively stable but never statistically significant

5. Interpretation

Why Might Compound Shocks Be Less Damaging?

Several mechanisms could explain the pattern:

1. **Diminishing marginal destruction**: The first shock destroys the most vulnerable capital stock. A second shock has less to destroy — the remaining infrastructure was either rebuilt to a higher standard or was resilient enough to survive the first shock.

2. **Preparedness and learning**: After a shock, governments and households invest in preparedness (early warning, evacuation plans, building codes). A follow-up shock arrives into a better-prepared environment.

3. **Mechanical baseline effect**: The LP measures $y_{i,t+h} - y_{i,t-1}$. For compound shocks, $y_{i,t-1}$ is already depressed by the prior shock, so the measured decline from the (lower) baseline is mechanically smaller.

4. **Selection on intensity**: Countries that are hit every year (e.g., CHN, PHL) may experience weaker individual storms on average, while isolated shocks may be rarer but more extreme events.

Disentangling the Mechanisms

- The **continuous intensity** robustness check (Slide 12) is key: when we control for wind speed continuously, the compound/isolated gap disappears. This suggests the binary result reflects **intensity selection**, not a true interaction effect.
- The **mechanical baseline** concern is partially addressed by the LP design — $y_{i,t-1}$ is already below trend for compound shocks, so we are measuring the additional damage from this lower starting point. The country trends absorb the systematic component.
- The **diminishing marginal** and **preparedness** stories are hard to separate without structural modeling, but both predict the observed pattern.

Summary and Takeaways

1. **Isolated shocks appear more damaging** than compound shocks (-3.0% vs -1.3% at $h = 5$ with p95), but the difference is **never statistically significant**.
2. The pattern is qualitatively **stable across thresholds** (p95 through Cat 2+), FE specifications, and sample restrictions. At very high thresholds (Cat 3+), effects weaken due to power.
3. With **continuous intensity**, the gap disappears — intensity selection (frequently-hit countries experience weaker storms on average) likely drives much of the binary result.
4. **The threshold matters**: The baseline p95 (60 knots) is very low, making countries like AUS and CHN "treated" nearly every year. Higher thresholds produce more within-country variation but fewer observations. Cat 2+ (83 knots) offers a reasonable trade-off.
5. **No evidence of compounding damage**: Repeated shocks do not amplify GDP losses. Statistical power is limited — only 19 countries have both types at p95 (fewer at higher thresholds).